OEIS A391810: formula (5), COMMENTS (C.1)–(C.10), and prime-index vanishing

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Abstract

We prove formula (5), COMMENTS (C.1)–(C.10), and the zero clause of COMMENT (C.11) of Paul D. Hanna's OEIS entry A391810, revision 39 [1]. For each positive integer n, let $T(n,k)$ be the coefficient of x^{-k} in the formal bilateral Laurent series $\sum_{j\in\mathbb{Z}} x^j(1-x^j)^{n+j}$ at zero. We prove that the negative alternating sum of the complete row is $2\,\mathrm{round}(2^n/3) = 2J_n$, where J_n is the offset-zero Jacobsthal sequence. The proof identifies every nonpositive Laurent coefficient with a coefficient of a finite polynomial, proves the stated row bound, and evaluates that polynomial at -1. No substitution into the bilateral series is required. The same polynomial proves the conjectured row sums, first and second moments, and alternating first-moment generating function in formulas (2), (3), (4), and (6) of that entry. A divisor-sum expression gives the individual terms and shows that $T(n,q) = 0$ for every prime $q > n$. The further assertion in (C.11) that the next prime is the first zero remains open here.

1 The identity and its conventions

Formula (5) and COMMENTS (C.1)–(C.11) of Paul D. Hanna's OEIS entry A391810 pose alternating-sum and individual-term conjectures [1]. The negative alternating sum of the full signed Laurent row equals twice a Jacobsthal number. We establish this identity and the ten individual-term formulas, together with prime-index vanishing, using the original row interpretation and inclusive endpoint. The Jacobsthal closed form and nearest-integer description are already recorded in A001045 [2]; the issue here is their connection to the bilateral Laurent coefficients. The pinned formal source [4] supplies the same connection and the whole formula. This note gives a self-contained written proof.

For $n \geq 1$, work in the formal Laurent field $\mathbb{Q}((x))$ and put

$$S_n(x) = \sum_{j\in\mathbb{Z}} x^j(1-x^j)^{n+j}, \qquad T(n,k) = [x^{-k}]S_n(x), \quad k \geq 0. \tag{1}$$

Integer powers with negative exponents mean inverses in this field. The $j = 0$ term is zero, since $n > 0$. We justify the coefficientwise sum below; it is not an analytic convergence assertion. Write

$$B_n = \left\lfloor \frac{n+1}{2} \right\rfloor^2, \qquad J_0 = 0, \quad J_1 = 1, \quad J_{n+2} = J_{n+1} + 2J_n.$$

For rational arguments, round denotes nearest-integer rounding, with ties toward positive infinity. The arguments used here are never ties.

Theorem 1. *For every integer $n \geq 1$, all $T(n,k)$ are integers, $T(n,k) = 0$ for $k > B_n$, and*

$$-\sum_{k=0}^{B_n}(-1)^k T(n,k) = 2\,\mathrm{round}\left(\frac{2^n}{3}\right) = 2J_n. \tag{2}$$

2 From the bilateral series to a finite row

Lemma 1. *For every $n \geq 1$ and $k \geq 0$,*

$$T(n,k) = [y^k]P_n(y), \qquad P_n(y) = -1 + \sum_{h=1}^{n} y^h(1-y^h)^{n-h} \in \mathbb{Z}[y]. \tag{3}$$

Moreover, $\deg P_n \leq B_n$.

Proof. First consider the formal sum in (1). For $j > 0$, its jth summand has least exponent j. For $j = -h < 0$, the identity $1 - x^{-h} = -x^{-h}(1-x^h)$ gives

$$x^{-h}(1-x^{-h})^{n-h} = (-1)^{n-h}x^{h(h-n-1)}(1-x^h)^{n-h}. \tag{4}$$

The last factor has constant coefficient one and only nonnegative exponents, also when $n-h < 0$. Thus the least exponent is $h(h-n-1)$. These least exponents tend to positive infinity as $h \to \infty$; the positive-index least exponents do likewise. They are uniformly bounded below, and only finitely many summands can contribute to any fixed coefficient. This proves that the bilateral coefficientwise sum is a well-defined element of $\mathbb{Q}((x))$.

No $j > 0$ term contributes to a nonpositive power of x. For $1 \leq h \leq n$, the $j = -h$ term is the finite polynomial $y^h(1-y^h)^{n-h}$ in $y = x^{-1}$. At the boundary $h = n+1$, it is

$$x^{-(n+1)}(1-x^{-(n+1)})^{-1} = -\frac{1}{1-x^{n+1}} = -\sum_{r\geq 0} x^{(n+1)r}.$$

Its nonpositive part is exactly the constant -1. For $h \geq n+2$, the least exponent $h(h-n-1)$ is positive. This proves (3), including the constant coefficient.

It remains to check the sharper row bound. The hth summand of $P_n + 1$ has degree $h(n+1-h)$. If $n = 2m+1$, this is at most $(m+1)^2 = B_n$, as follows by completing the square. If $n = 2m$, split off its highest monomial

$$(-1)^{2m-h}y^{h(2m+1-h)}, \qquad 1 \leq h \leq 2m.$$

Pairing h with $2m+1-h$ cancels these monomials: their exponents coincide and their signs are opposite. There is no fixed index in this pairing. Every remaining monomial in the hth summand has exponent $h(r+1)$ with $0 \leq r \leq 2m-h-1$, hence at most $h(2m-h) \leq m^2 = B_n$. When that range is empty, there is no remaining monomial. The constant term also respects the bound. This proves the lemma. □

3 Polynomial evaluation and Jacobsthal numbers

Proof of Theorem 1. By Lemma 1, the full sum in (2) is $-P_n(-1)$. We evaluate the finite polynomial, not S_n. An odd index h contributes -2^{n-h} to $P_n(-1)$. An even index $h < n$ contributes zero; if $h = n$ is even, its contribution is one, since its exponent $n-h$ is zero. For $n = 2m+1$ with $m \geq 0$, the odd indices give

$$P_{2m+1}(-1) = -1 - \sum_{r=0}^{m} 4^r = -\frac{4^{m+1}+2}{3} = -\frac{2(2^{2m+1}+1)}{3}.$$

For $n = 2m$ with $m \geq 1$, the even boundary term cancels the constant -1, so

$$P_{2m}(-1) = -2\sum_{r=0}^{m-1} 4^r = -\frac{2(4^m - 1)}{3}.$$

Together these yield

$$-P_n(-1) = \frac{2(2^n - (-1)^n)}{3}. \tag{5}$$

To identify the right-hand side, put $C_n = (2^n - (-1)^n)/3$. Then $C_0 = 0$, $C_1 = 1$, and a direct calculation gives

$$C_{n+1} + 2C_n = \frac{2^{n+2} - ((-1)^{n+1} + 2(-1)^n)}{3} = \frac{2^{n+2} - (-1)^{n+2}}{3} = C_{n+2}.$$

Induction in the defining recurrence therefore gives $C_n = J_n$ for all $n \geq 0$. In particular,

$$\frac{2^n}{3} - J_n = \frac{(-1)^n}{3}.$$

The absolute error is $1/3 < 1/2$, so the unique nearest integer to $2^n/3$ is J_n. Combining this with (5) proves both equalities. Integrality and the vanishing assertion follow from Lemma 1. □

The first six alternating sums are $2, 2, 6, 10, 22, 42$. The positive-index restriction matters: the finite expression gives $P_0(y) = -1$, so its negative value at -1 is 1, whereas $2J_0 = 0$. No zero-index row extension is asserted.

4 Individual terms and prime-index zeros

We use $\binom{a}{b} = 0$ when $a \geq 0$ and b is outside $0 \leq b \leq a$.

Corollary 1 (Divisor formula)**.** *For $n \geq 1$, $T(n, 0) = -1$, and for every $k \geq 1$,*

$$T(n,k) = \sum_{\substack{h \mid k \\ 1 \leq h \leq n}} (-1)^{k/h-1} \binom{n-h}{k/h-1}. \tag{6}$$

Proof. In the hth summand of $P_n + 1$, the binomial expansion is

$$y^h(1-y^h)^{n-h} = \sum_{r=0}^{n-h} (-1)^r \binom{n-h}{r} y^{h(r+1)}.$$

Its y^k coefficient is zero unless $h \mid k$; otherwise put $r = k/h - 1$. All these summands have positive degree, so the constant coefficient of P_n is -1. □

Corollary 2 (COMMENTS (C.1)–(C.5))**.** *The first five individual-term conjectures hold in their stated ranges:*

$$T(n,0) = -1 \quad (n \geq 1), \qquad T(n,1) = 1 \quad (n \geq 1),$$

$$T(n,2) = -(n-2) \quad (n \geq 2), \qquad T(n,3) = \frac{n(n-3)}{2} + 2 \quad (n \geq 3),$$

$$T(n,4) = -\frac{(n-3)(n^2-3n+8)}{6} \quad (n \geq 4).$$

Proof. The constant term is given above. For $k = 1, 2, 3, 4$, respectively, formula (6) gives

$$1, \qquad -\binom{n-1}{1} + 1, \qquad \binom{n-1}{2} + 1, \qquad -\binom{n-1}{3} - \binom{n-2}{1} + 1.$$

The divisors involved are $\{1\}$, $\{1, 2\}$, $\{1, 3\}$, and $\{1, 2, 4\}$. Expanding the displayed binomials gives the stated formulas. □

Corollary 3 (COMMENTS (C.6)–(C.9))**.** *For $m \geq 1$,*

$$T(2m-1, m^2) = (-1)^{m-1}, \qquad T(2m, m^2) = (-1)^{m-1}m.$$

For $m \geq 2$,

$$T(2m-1, m^2-1) = 2(-1)^m, \qquad T(2m, m^2-1) = 2m(-1)^m.$$

Proof. For the odd row, the hth summand has degree $h(2m-h) = m^2-(h-m)^2$. Only $h = m$ reaches m^2; its highest coefficient is $(-1)^{m-1}$. To reach $m^2 - 1$, only $h = m-1, m, m+1$ are possible. The middle index contributes nothing, because m does not divide $m^2 - 1$ when $m \geq 2$. The other two contribute their highest coefficients, both $(-1)^m$.

For the even row, first remove the highest monomial of each summand. Their total is zero by the pairing used in Lemma 1. The remaining hth polynomial has degree at most $h(2m-h) = m^2 - (h-m)^2$; when nonempty, its highest coefficient is

$$(-1)^{2m-h-1}\binom{2m-h}{2m-h-1} = (-1)^{2m-h-1}(2m-h).$$

Thus $h = m$ alone contributes to m^2, with coefficient $(-1)^{m-1}m$. For $m^2 - 1$ and $m \geq 2$, only $h = m-1, m+1$ contribute, since the same divisibility argument excludes $h = m$. Their contributions are $(-1)^m(m+1)$ and $(-1)^m(m-1)$. □

Corollary 4 (COMMENT (C.10) and the zero clause of (C.11))**.** *For every odd prime p, $T(p, p) = 2$. For every $n \geq 1$ and prime $q > n$, $T(n, q) = 0$. In particular, the next prime after n has zero coefficient, as asserted for $n \geq 5$ in (C.11).*

Proof. For $T(p, p)$ the divisors in (6) are $1, p$. Since p is odd their contributions are

$$(-1)^{p-1}\binom{p-1}{p-1} = 1 \qquad \text{and} \qquad \binom{0}{0} = 1.$$

If $q > n$, only the divisor 1 is allowed, and $\binom{n-1}{q-1} = 0$. □

The stronger assertion in (C.11) is that $T(n, k) \neq 0$ for every $0 \leq k < q$, where q is the next prime after $n \geq 5$. That first-zero assertion is not established by the divisor formula; it remains an open question in this note. Exact checks of rows $5 \leq n \leq 79$ agree with it.

5 The sibling conjectures

All sums below retain the complete inclusive endpoint B_n. Let $D = y\,d/dy$ denote the Euler differential operator.

Corollary 5. *Formulas (2) and (3) of A391810 hold for every $n \geq 1$:*

$$\sum_{k=0}^{B_n} T(n,k) = 0, \qquad \sum_{k=0}^{B_n} kT(n,k) = 1.$$

Formula (4) holds for every $n \geq 2$:

$$\sum_{k=0}^{B_n} k^2 T(n,k) = 5 - 2n.$$

Proof. At $y = 1$, only the $h = n$ summand of $P_n + 1$ survives, giving $P_n(1) = 0$. In $P_n'(1)$, the terms $h = n, n-1$ give $n - (n-1) = 1$; for $n = 1$, use $P_1(y) = -1 + y$ directly. For $D^2 P_n(1)$, only $n - h = 0, 1, 2$ survive. Applying $D^2 y^a = a^2 y^a$ to y^h, $y^h - y^{2h}$ and $y^h - 2y^{2h} + y^{3h}$ gives

$$(D^2 P_n)(1) = n^2 - 3(n-1)^2 + 2(n-2)^2 = 5 - 2n.$$

When $n = 2$, the absent $h = 0$ term has the displayed weight zero. Lemma 1 identifies these three evaluations with the stated sums. □

Corollary 6. *Formula (6) of A391810 holds. Namely, for*

$$A_n = -\sum_{k=0}^{B_n} (-1)^k kT(n,k) = P_n'(-1), \qquad n \geq 1,$$

the formal ordinary generating function is

$$\sum_{n \geq 1} A_n z^n = \frac{z(1 - z + 8z^2)}{(1+z)^2 (1-2z)^2}. \tag{7}$$

Proof. Write $n = h + r$ and $f_{h,r}(y) = y^h (1 - y^h)^r$. For odd h, direct differentiation gives $f_{h,r}'(-1) = h(2^r + r2^{r-1})$, with the second term zero when $r = 0$. For even h, the derivatives for $r = 0, 1$ are $-h, h$, and all others vanish. Consequently,

$$\sum_{r \geq 0} z^{h+r} f_{h,r}'(-1) = \begin{cases} hz^h (1-z)/(1-2z)^2, & h \text{ odd}, \\ hz^h (z-1), & h \text{ even}. \end{cases}$$

Each coefficient involves finitely many h, so rearrangement in $\mathbb{Q}[[z]]$ is valid. Differentiating the geometric series gives

$$\sum_{h \text{ odd}} hz^h = \frac{z(1+z^2)}{(1-z^2)^2}, \qquad \sum_{h \text{ even}} hz^h = \frac{2z^2}{(1-z^2)^2}.$$

These identities yield

$$\sum_{n \geq 1} A_n z^n = \frac{1-z}{(1-z^2)^2} \left(\frac{z(1+z^2)}{(1-2z)^2} - 2z^2 \right) = \frac{z(1 - z + 8z^2)}{(1+z)^2 (1-2z)^2}.$$

□

The first eight A_n are $1, 1, 13, 25, 81, 181, 453, 1025$, as in the source. Formula (1) defines the row. Formula (7), which is not labeled a conjecture there, is the defining squared-norm relation for the companion entry A391811 [3]. Its half-sum of squares is an integer: $T(n,k)^2 \equiv T(n,k) \pmod 2$ and the row sum is zero. It is positive, since $T(n,0) = -1$ and $T(n,1) = 1$. The row-sum COMMENT in A391811 is also Corollary 5.

Coverage of the posing entry. The source locators in the following table refer to revision 39 of A391810.

Source item	Mathematical coverage
Formula (1); row endpoint	Definition; Lemma 1
Formulas (2), (3), (4)	Corollary 5
Formula (5)	Theorem 1
Formula (6)	Corollary 6
COMMENTS (C.1)–(C.5)	Corollary 2
COMMENTS (C.6)–(C.9)	Corollary 3
COMMENT (C.10)	Corollary 4
(C.11), next-prime coefficient zero	Corollary 4, for every prime $q > n$
(C.11), first-zero location	Open here; finite checks for $5 \le n \le 79$
"Each row sum appears to equal zero"	Same statement as formula (2)
Formula (7); A391811	Defining relation; positive integrality as above
Two related bilateral identities in COMMENTS	Outside the nonpositive-coefficient results here

Source and verification scope. The result is stated as `full_formula_five` in the pinned module in [4]. Its five clauses retain the integer row sum, the rounding–Jacobsthal equality, the rational original-Laurent sum, the coefficient bridge for every $k \ge 0$, and vanishing beyond B_n. The companion `HannaLaurentRowSums.lean` contains `series_coefficient`, `row_degree`, and `result`: the coefficient bridge, support bound, and formula (2). The declarations `result` in `HannaLaurentFirstMoment.lean` and `HannaLaurentSecondMoment.lean` state formulas (3) and (4). All four modules have the revision specified in [4]. The divisor formula, COMMENTS (C.1)–(C.10), prime-index vanishing, and formula (6) have the written proofs above.

The companion script `verify.py`, with results in `verification.json`, compares the divisor formula and an independent Laurent expansion with the polynomial on full rows $1 \le n \le 79$, checks the term formulas through row 79, and checks all four moment identities. The Jacobsthal scalar identity is checked through $n = 512$. These are finite exact-arithmetic checks supplementing the written proofs.

References

[1] P. D. Hanna, *Entry A391810*, The On-Line Encyclopedia of Integer Sequences, revision 39, 3 February 2026; author line 21 December 2025. Definition, row endpoint, formulas (2)–(6), and COMMENTS (C.1)–(C.11). https://oeis.org/A391810.

[2] The OEIS Foundation Inc., *A001045: Jacobsthal sequence*, The On-Line Encyclopedia of Integer Sequences, revision 1350, 30 June 2026. Name, offset, recurrence, and closed-form/rounding formulas. https://oeis.org/A001045.

[3] P. D. Hanna, *Entry A391811*, The On-Line Encyclopedia of Integer Sequences, revision 11, 28 December 2025; author line 21 December 2025. Definition and formula for the half-sum of squared A391810 row entries; row-sum COMMENT. https://oeis.org/A391811.

[4] Trureturing contributors, *Formal sources for A391810*, directory `D5/S3/ArithSums`, revision `1239bd2a0ed6e9d54aabb5a42af94234e9eb61ae`. `HannaLaurentAlternatingJacobsthal.lean`, theorem `full_formula_five`; companion modules linked above.